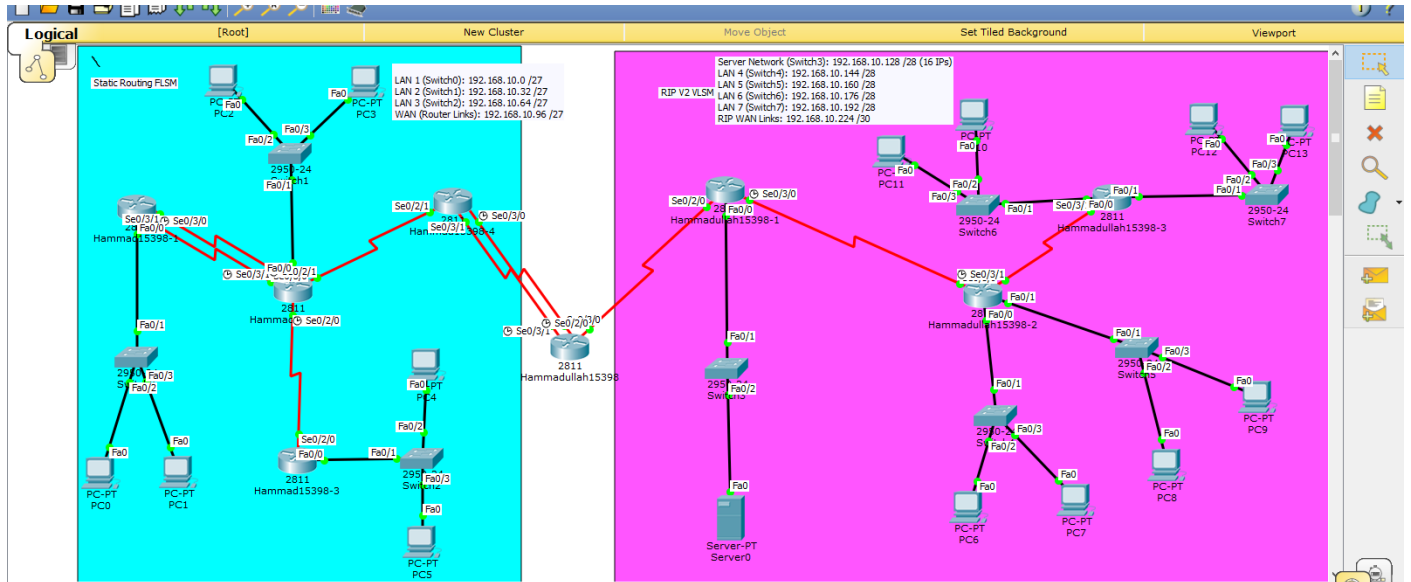
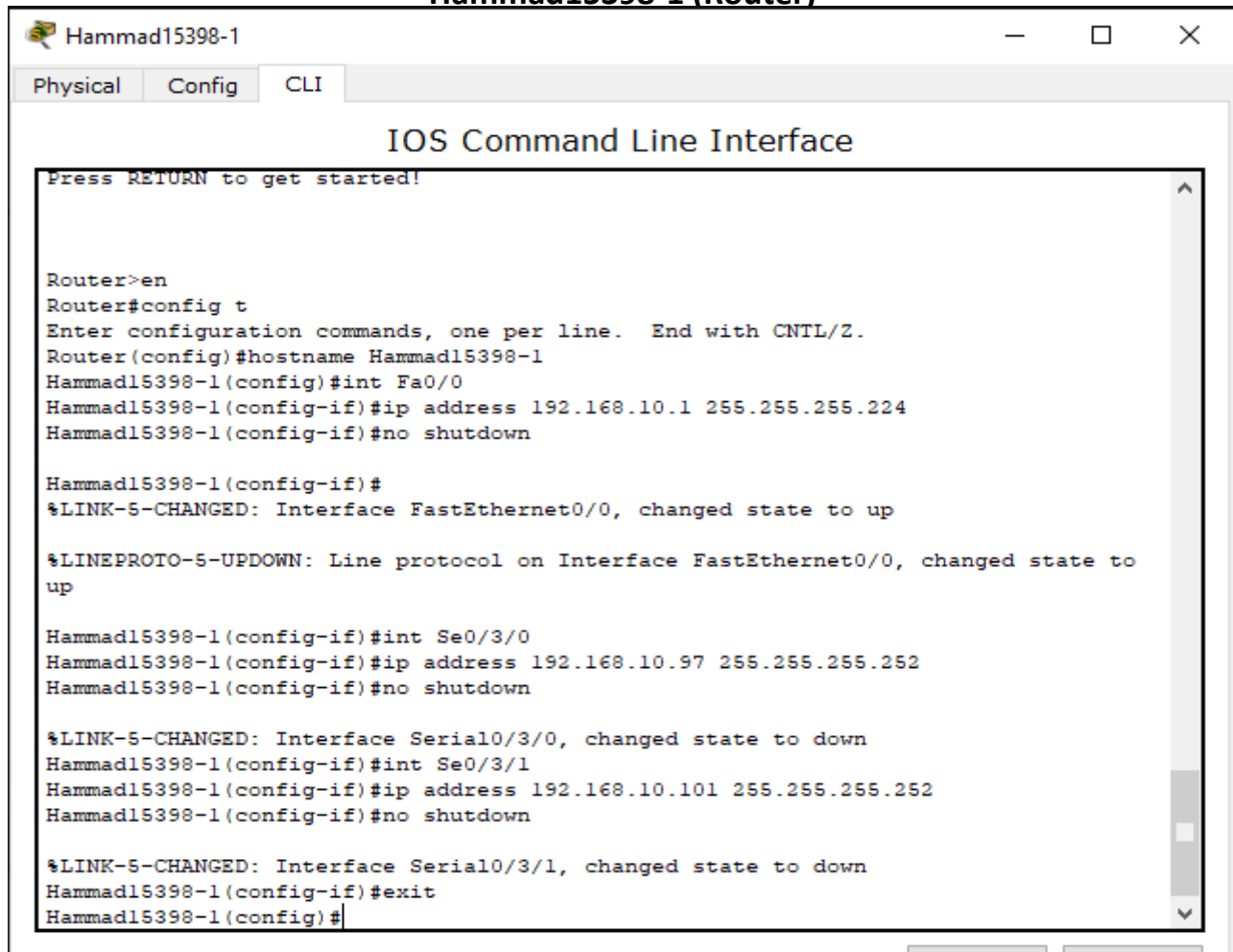


Topology Diagram:



Hammad15398-1 (Router)



```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Hammad15398-1
Hammad15398-1(config)#int Fa0/0
Hammad15398-1(config-if)#ip address 192.168.10.1 255.255.255.224
Hammad15398-1(config-if)#no shutdown

Hammad15398-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

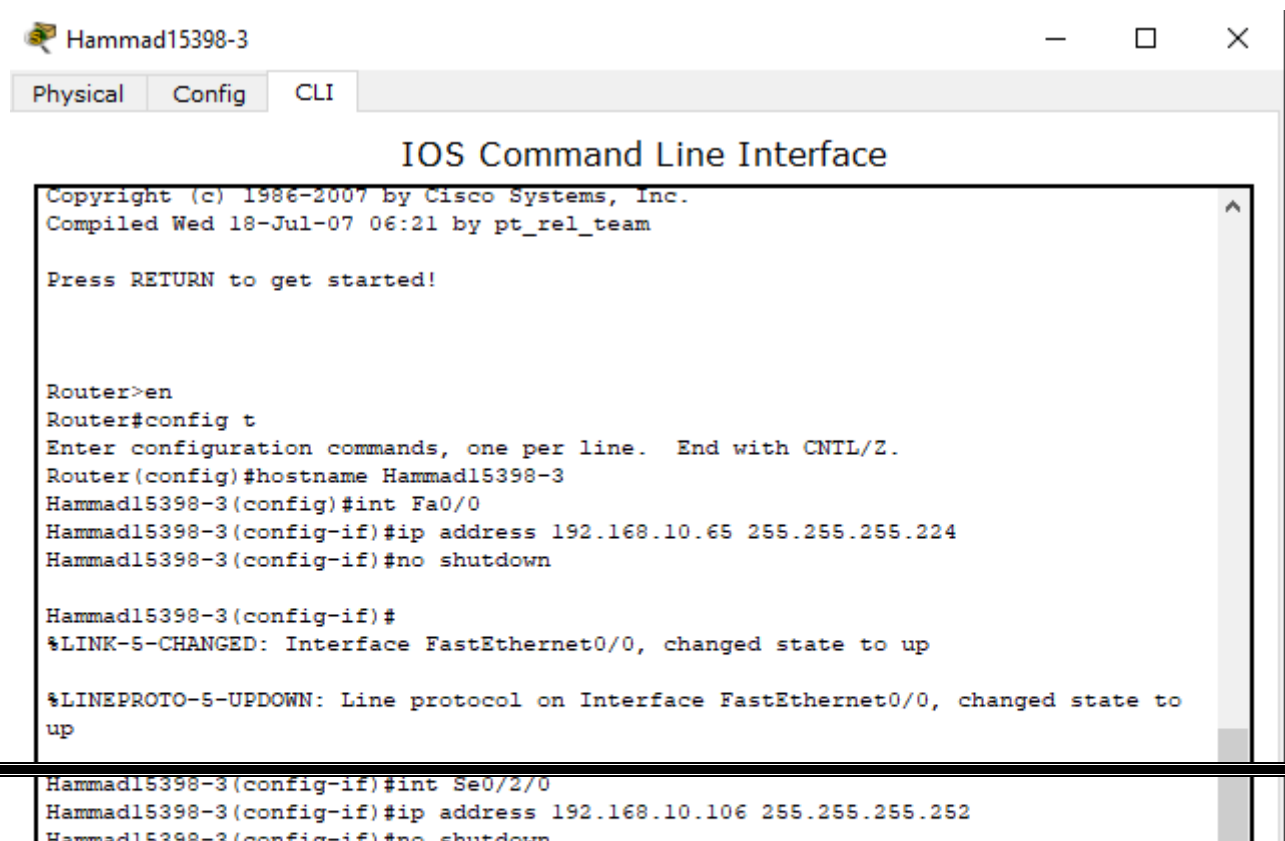
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Hammad15398-1(config-if)#int Se0/3/0
Hammad15398-1(config-if)#ip address 192.168.10.97 255.255.255.252
Hammad15398-1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Hammad15398-1(config-if)#int Se0/3/1
Hammad15398-1(config-if)#ip address 192.168.10.101 255.255.255.252
Hammad15398-1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
Hammad15398-1(config-if)#exit
Hammad15398-1(config)#
```

Hammad15398-3 (Router)



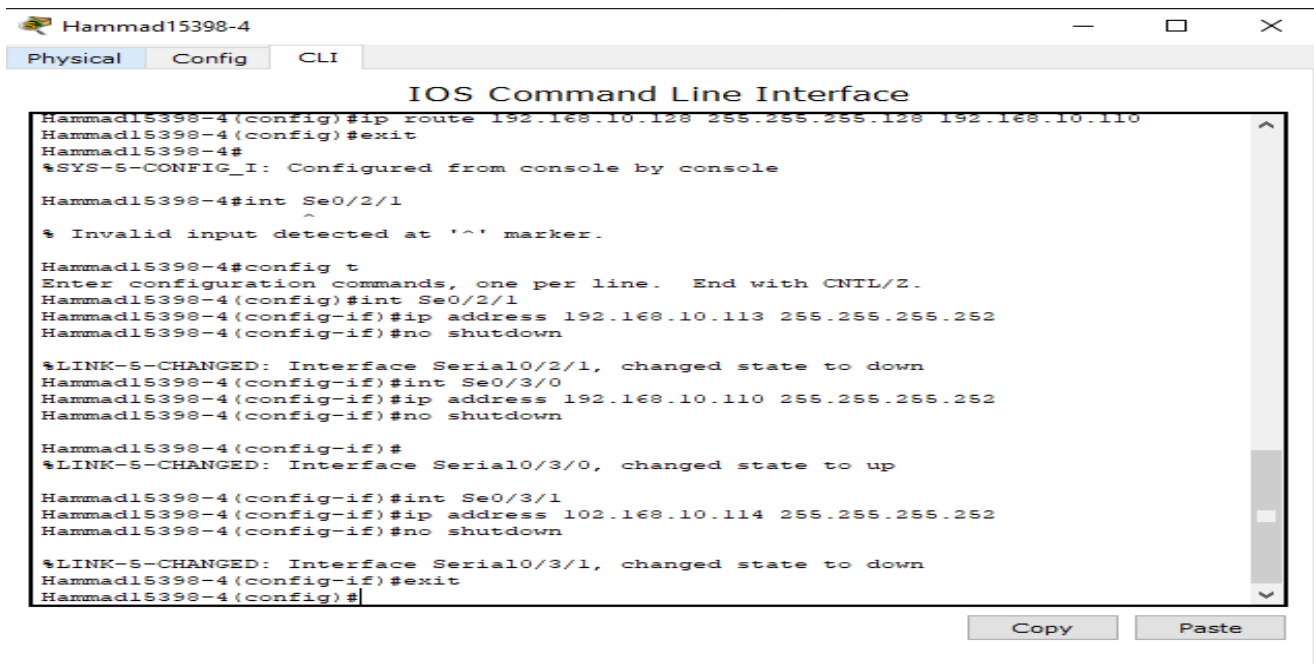
```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Hammad15398-3
Hammad15398-3(config)#int Fa0/0
Hammad15398-3(config-if)#ip address 192.168.10.65 255.255.255.224
Hammad15398-3(config-if)#no shutdown

Hammad15398-3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Hammad15398-3(config-if)#int Se0/2/0
Hammad15398-3(config-if)#ip address 192.168.10.106 255.255.255.252
Hammad15398-3(config-if)#no shutdown
```

Hammad15398-4



```
Hammad15398-4
Physical Config CLI
IOS Command Line Interface
Hammad15398-4(config)#ip route 192.168.10.128 255.255.255.128 192.168.10.110
Hammad15398-4(config)#exit
Hammad15398-4#
%SYS-5-CONFIG_I: Configured from console by console
Hammad15398-4#int Se0/2/1
^
% Invalid input detected at '^' marker.
Hammad15398-4#config t
Enter configuration commands, one per line. End with CNTL/Z.
Hammad15398-4(config)#int Se0/2/1
Hammad15398-4(config-if)#ip address 192.168.10.113 255.255.255.252
Hammad15398-4(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/2/1, changed state to down
Hammad15398-4(config-if)#int Se0/3/0
Hammad15398-4(config-if)#ip address 192.168.10.110 255.255.255.252
Hammad15398-4(config-if)#no shutdown
Hammad15398-4(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
Hammad15398-4(config-if)#int Se0/3/1
Hammad15398-4(config-if)#ip address 102.168.10.114 255.255.255.252
Hammad15398-4(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
Hammad15398-4(config-if)#exit
Hammad15398-4(config)#
```

Copy Paste

CID: 119638

Hammad15398-2 (Router)

 Hammad15398-2

Physical Config CLI

```
239K bytes of non-volatile configuration memory.
62720K bytes of ATA CompactFlash (Read/Write)
Cisco IOS Software, 2800 Software (C2800NM-ADVIPSERVICESK9-M), Version 12.4(15)T1, RELEASE SOF
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 18-Jul-07 06:21 by pt_rel_team
```

Press RETURN to get started!

```
Hammad15398-2>en
Hammad15398-2#config t
Enter configuration commands, one per line. End with CNTL/Z.
Hammad15398-2(config)#int Fa0/0
Hammad15398-2(config-if)#ip address 192.168.10.33 255.255.255.224
Hammad15398-2(config-if)#no shutdown

Hammad15398-2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Hammad15398-2(config-if)#int Se0/3/0
Hammad15398-2(config-if)#ip address 192.168.10.98 255.255.255.252
Hammad15398-2(config-if)#no shutdown

Hammad15398-2(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

Hammad15398-2(config-if)#int Se0/3/1
Hammad15398-2(config-if)#ip address 192.168.10.102 255.255.255.252
Hammad15398-2(config-if)#no shutdown

Hammad15398-2(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

Hammad15398-2(config-if)#int Se0/2/0
Hammad15398-2(config-if)#ip address 192.168.10.105 255.255.255.252
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

^
% Invalid input detected at '^' marker.

Hammad15398-2(config-if)#int Se0/2/0
Hammad15398-2(config-if)#ip address 192.168.10.105 255.255.255.252
Hammad15398-2(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/2/0, changed state to down
Hammad15398-2(config-if)#exit
Hammad15398-2(config)#
```

CID: 119638

Hammadullah15398-1 (Router)

Hammadullah15398-1

Physical Config CLI

IOS Command Line Interface

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Hammadullah15398-1
Hammadullah15398-1(config)#int Fa0/0
Hammadullah15398-1(config-if)#ip address 192.168.10.129 255.255.255.240
Hammadullah15398-1(config-if)#no shutdown

Hammadullah15398-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Hammadullah15398-1(config-if)#int Se0/2/0
Hammadullah15398-1(config-if)#ip address 192.168.10.225 255.255.255.252
Hammadullah15398-1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/2/0, changed state to down
Hammadullah15398-1(config-if)#exit
Hammadullah15398-1(config)#int Se0/3/0
Hammadullah15398-1(config-if)#ip address 192.168.10.229 255.255.255.252
Hammadullah15398-1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Hammadullah15398-1(config-if)#exit
Hammadullah15398-1(config)#
```

Copy

Paste

Server0**ON Next page**

Server0

Physical Config Services Desktop Custom Interface

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IP Address: 192.168.10.130

Subnet Mask: 255.255.255.240

Default Gateway: 192.168.10.129

DNS Server:

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address: /

Link Local Address: FE80::201:96FF:FE4E:9EEA

IPv6 Gateway:

IPv6 DNS Server:

Server0

Physical Config Services Desktop Custom Interface

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: LAN4

Default Gateway: 192.168.10.145

DNS Server: 0.0.0.0

Start IP Address : 192 168 10 146

Subnet Mask: 255 255 255 240

Maximum number of Users : 14

TFTP Server: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server
LAN7	192.168.10.193	0.0.0.0	192.168.10.193	255.255.255.240	14	0.0.0.0
LAN6	192.168.10.177	0.0.0.0	192.168.10.177	255.255.255.240	14	0.0.0.0
LAN5	192.168.10.161	0.0.0.0	192.168.10.161	255.255.255.240	14	0.0.0.0
LAN4	192.168.10.145	0.0.0.0	192.168.10.145	255.255.255.240	14	0.0.0.0
server0	0.0.0.0	0.0.0.0	192.168.10.1	255.255.255.240	128	0.0.0.0

Server0

Physical Config **Services** Desktop Custom Interface

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: LAN5

Default Gateway: 192.168.10.161

DNS Server: 0.0.0.0

Start IP Address : 192 168 10 162

Subnet Mask: 255 255 255 240

Maximum number of Users : 14

TFTP Server: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	Time
LAN7	192.168.10.193	0.0.0.0	192.168.10....	255.255....	14	0.0.0
LAN6	192.168.10.177	0.0.0.0	192.168.10....	255.255....	14	0.0.0
LAN5	192.168.10.161	0.0.0.0	192.168.10....	255.255....	14	0.0.0
LAN4	192.168.10.145	0.0.0.0	192.168.10....	255.255....	14	0.0.0

Server0

Physical Config **Services** Desktop Custom Interface

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: LAN6

Default Gateway: 192.168.10.177

DNS Server: 0.0.0.0

Start IP Address : 192 168 10 178

Subnet Mask: 255 255 255 240

Maximum number of Users : 14

TFTP Server: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	Time
LAN7	192.168.10.193	0.0.0.0	192.168.10....	255.255....	14	0.0.0
LAN6	192.168.10.177	0.0.0.0	192.168.10....	255.255....	14	0.0.0
LAN5	192.168.10.161	0.0.0.0	192.168.10....	255.255....	14	0.0.0

Server0

Physical Config Services Desktop Custom Interface

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: LAN7

Default Gateway: 192.168.10.193

DNS Server: 0.0.0.0

Start IP Address : 192 168 10 194

Subnet Mask: 255 255 255 240

Maximum number of Users : 14

TFTP Server: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server
LAN7	192.168.10.193	0.0.0.0	192.168.10.194	255.255.255.240	14	0.0.0.0
LAN6	192.168.10.177	0.0.0.0	192.168.10.178	255.255.255.240	14	0.0.0.0
LAN5	192.168.10.161	0.0.0.0	192.168.10.162	255.255.255.240	14	0.0.0.0
LAN4	192.168.10.145	0.0.0.0	192.168.10.146	255.255.255.240	14	0.0.0.0

Hammadullah15398-1

Hammadullah15398-1

Physical Config CLI

IOS Command Line Interface

```

Hammadullah15398-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to
up

Hammadullah15398-1(config-if)#int Se0/2/0
Hammadullah15398-1(config-if)#ip address 192.168.10.225 255.255.255.252
Hammadullah15398-1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/2/0, changed state to down
Hammadullah15398-1(config-if)#exit
Hammadullah15398-1(config)#int Se0/3/0
Hammadullah15398-1(config-if)#ip address 192.168.10.229 255.255.255.252
Hammadullah15398-1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Hammadullah15398-1(config-if)#exit
Hammadullah15398-1(config)#router rip
Hammadullah15398-1(config-router)#version 2
Hammadullah15398-1(config-router)#network 192.168.10.0
Hammadullah15398-1(config-router)#no auto-summary
Hammadullah15398-1(config-router)#passive interface Fa0/0
^
% Invalid input detected at '^' marker.

Hammadullah15398-1(config-router)#passive-interface Fa0/0
Hammadullah15398-1(config-router)#passive-interface Fa0/1
Hammadullah15398-1(config-router)#exit
Hammadullah15398-1(config)#
  
```


CID: 119638

Hammadullah15398-2

 Hammadullah15398-2

Physical Config CLI

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Hammadullah15398-2
Hammadullah15398-2(config)#router rip
Hammadullah15398-2(config-router)#version 2
Hammadullah15398-2(config-router)#network 192.168.10.0
Hammadullah15398-2(config-router)#no auto-summary
Hammadullah15398-2(config-router)#passive-interface Fa0/0
Hammadullah15398-2(config-router)#passive-interface Fa0/1
Hammadullah15398-2(config-router)#exit
Hammadullah15398-2(config)#int Fa0/0
Hammadullah15398-2(config-if)#ip helper-address 192.168.10.30
Hammadullah15398-2(config-if)#exit
Hammadullah15398-2(config)#int Fa0/1
Hammadullah15398-2(config-if)#ip helper-address 19
^
% Invalid input detected at '^' marker.

Hammadullah15398-2(config-if)#exit
Hammadullah15398-2(config)#interface FastEthernet0/0
Hammadullah15398-2(config-if)#
Hammadullah15398-2(config-if)#exit
Hammadullah15398-2(config)#interface FastEthernet0/1
Hammadullah15398-2(config-if)#
Hammadullah15398-2(config-if)#exit
Hammadullah15398-2(config)#interface FastEthernet0/0
Hammadullah15398-2(config-if)#exit
Hammadullah15398-2(config)#int Fa0/0
Hammadullah15398-2(config-if)#ip helper-address 192.168.10.130
Hammadullah15398-2(config-if)#int Fa0/1
Hammadullah15398-2(config-if)#ip helper-address 192.168.10.130
Hammadullah15398-2(config-if)#exit
Hammadullah15398-2(config)#int Se0/3/0
Hammadullah15398-2(config-if)#ip address 192.168.10.230 255.255.255.252
Hammadullah15398-2(config-if)#no shutdown

Hammadullah15398-2(config-if)#int
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
^
% Invalid input detected at '^' marker.

Hammadullah15398-2(config-if)#int Se0/3/1
Hammadullah15398-2(config-if)#ip address 192.168.10.233 255.255.255.252
Hammadullah15398-2(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
Hammadullah15398-2(config-if)#exit
Hammadullah15398-2(config)#
```

CID: 119638

Hammadullah15398-3

Hammadullah15398-3

Physical Config CLI

IOS Command Line Interface

Press RETURN to get started!

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Hammadullah15398-3
Hammadullah15398-3(config)#router rip
Hammadullah15398-3(config-router)#version 2
Hammadullah15398-3(config-router)#network 192.168.10.0
Hammadullah15398-3(config-router)#no auto-summary
Hammadullah15398-3(config-router)#passive-interface Fa0/0
Hammadullah15398-3(config-router)#passive-interface Fa0/1
Hammadullah15398-3(config-router)#exit
Hammadullah15398-3(config)#int Fa0/0
Hammadullah15398-3(config-if)#ip helper-address 192.168.10.130
Hammadullah15398-3(config-if)#int Fa0/1
Hammadullah15398-3(config-if)#ip helper-address 192.168.10.130
Hammadullah15398-3(config-if)#exit
Hammadullah15398-3(config)#int Se0/3/1
Hammadullah15398-3(config-if)#ip address 192.168.10.234 255.255.255.252
Hammadullah15398-3(config-if)#no shutdown

Hammadullah15398-3(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

Hammadullah15398-3(config-if)#exit
Hammadullah15398-3(config)#
```

Copy

Paste

CID: 119638

Hammad15398-1

Hammad15398-1

Physical Config CLI

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Hammad15398-1(config-if)#int Se0/3/0
Hammad15398-1(config-if)#ip address 192.168.10.97 255.255.255.252
Hammad15398-1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Hammad15398-1(config-if)#int Se0/3/1
Hammad15398-1(config-if)#ip address 192.168.10.101 255.255.255.252
Hammad15398-1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
Hammad15398-1(config-if)#exit
Hammad15398-1(config)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up

Hammad15398-1(config)#ip route 192.168.10.32 255.255.255.224 192.168.10.98
Hammad15398-1(config)#ip route 192.168.10.64 255.255.255.224 192.168.10.98
Hammad15398-1(config)#ip route 192.168.10.32 255.255.255.224 192.168.10.102 10
Hammad15398-1(config)#exit
Hammad15398-1#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy Paste

Hammad15398-4

Hammad15398-4

Physical Config CLI

IOS Command Line Interface

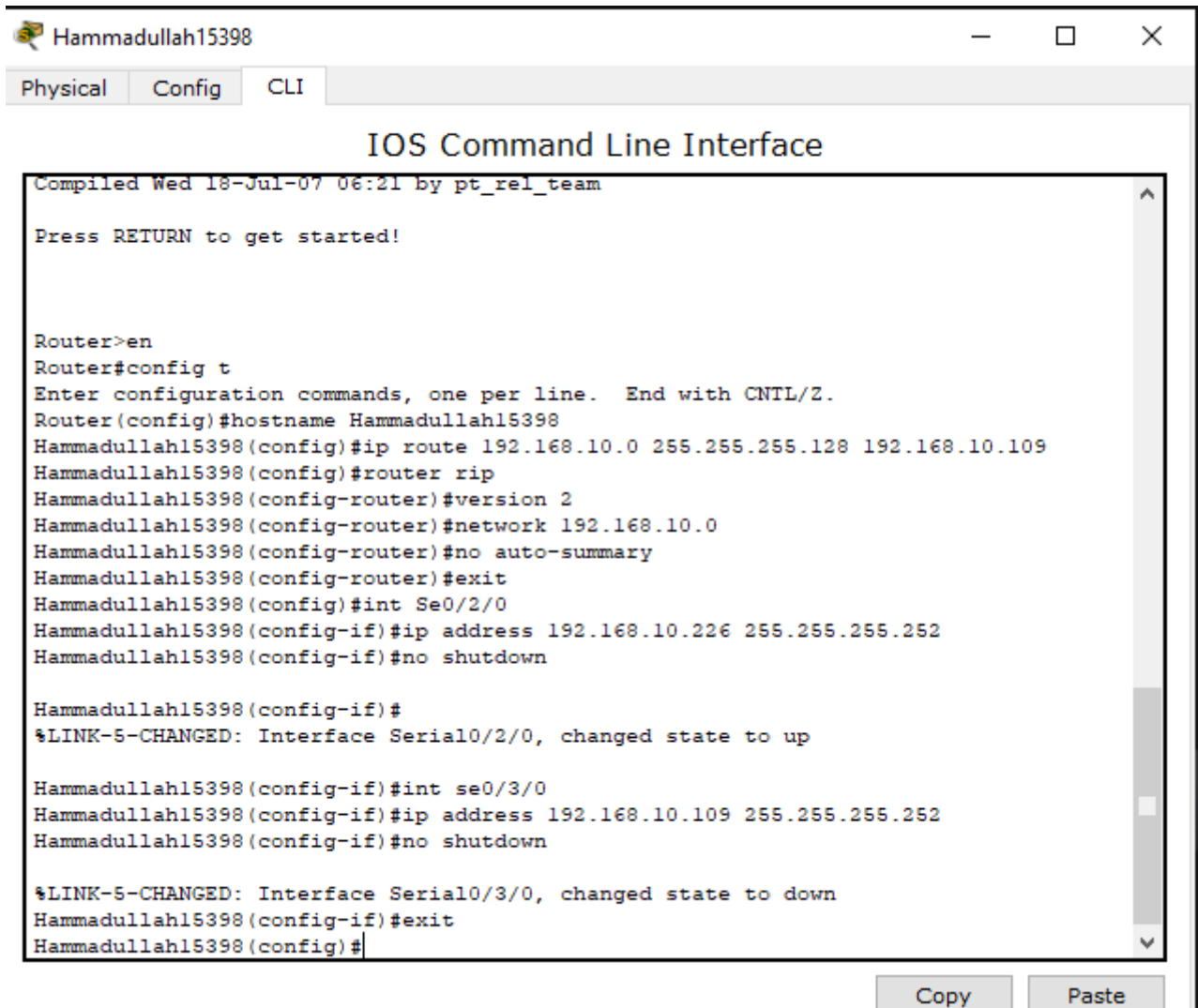
```
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.
cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
M860 processor: part number 0, mask 49
2 FastEthernet/IEEE 802.3 interface(s)
6 Low-speed serial(sync/async) network interface(s)
239K bytes of non-volatile configuration memory.
62720K bytes of ATA CompactFlash (Read/Write)
Cisco IOS Software, 2800 Software (C2800NM-ADVIPSERVICESK9-M), Version 12.4(15)T1,
RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 18-Jul-07 06:21 by pt_rel_team

Press RETURN to get started!

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Hammad15398-4
Hammad15398-4(config)#ip route 192.168.10.128 255.255.255.128 192.168.10.110
Hammad15398-4(config)#exit
Hammad15398-4#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy Paste



```
Compiled Wed 18-Jul-07 06:21 by pt_rel_team

Press RETURN to get started!

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Hammadullah15398
Hammadullah15398(config)#ip route 192.168.10.0 255.255.255.128 192.168.10.109
Hammadullah15398(config)#router rip
Hammadullah15398(config-router)#version 2
Hammadullah15398(config-router)#network 192.168.10.0
Hammadullah15398(config-router)#no auto-summary
Hammadullah15398(config-router)#exit
Hammadullah15398(config)#int Se0/2/0
Hammadullah15398(config-if)#ip address 192.168.10.226 255.255.255.252
Hammadullah15398(config-if)#no shutdown

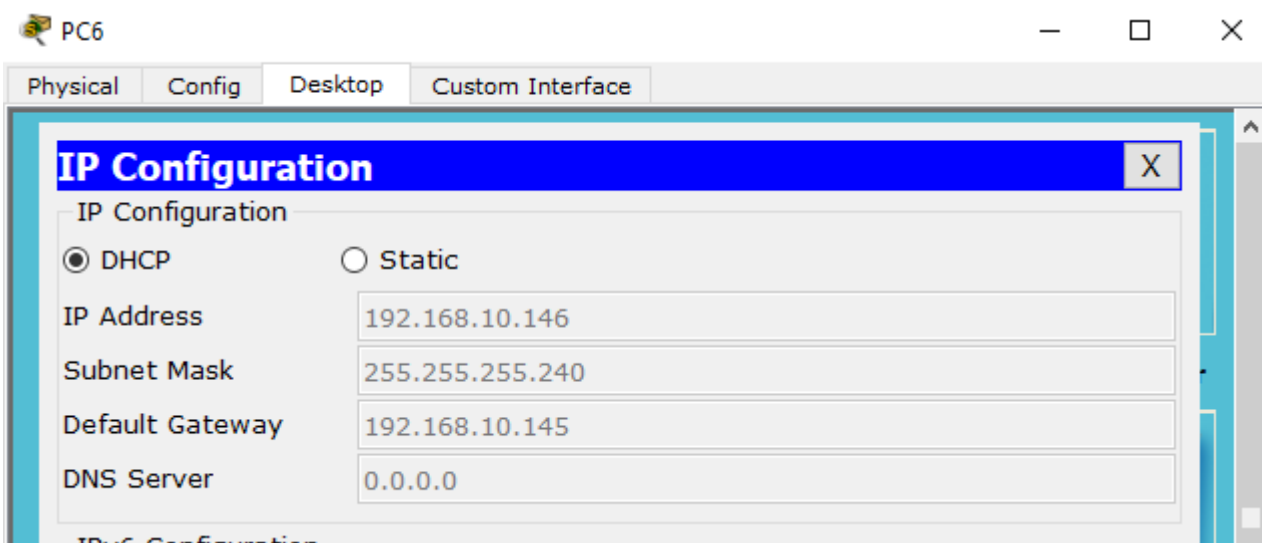
Hammadullah15398(config-if)#
%LINK-5-CHANGED: Interface Serial0/2/0, changed state to up

Hammadullah15398(config-if)#int se0/3/0
Hammadullah15398(config-if)#ip address 192.168.10.109 255.255.255.252
Hammadullah15398(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Hammadullah15398(config-if)#exit
Hammadullah15398(config)#
```

Copy Paste

PC6:



PC6

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

☒ DHCP ☐ Static

IP Address: 192.168.10.146

Subnet Mask: 255.255.255.240

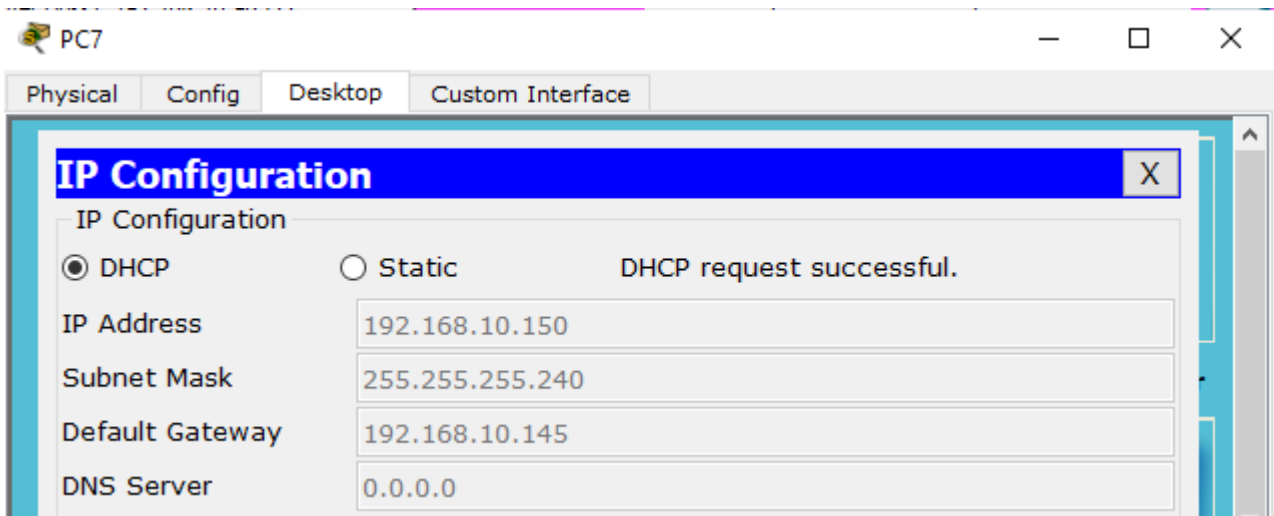
Default Gateway: 192.168.10.145

DNS Server: 0.0.0.0

IPv6 Configuration

CID: 119638

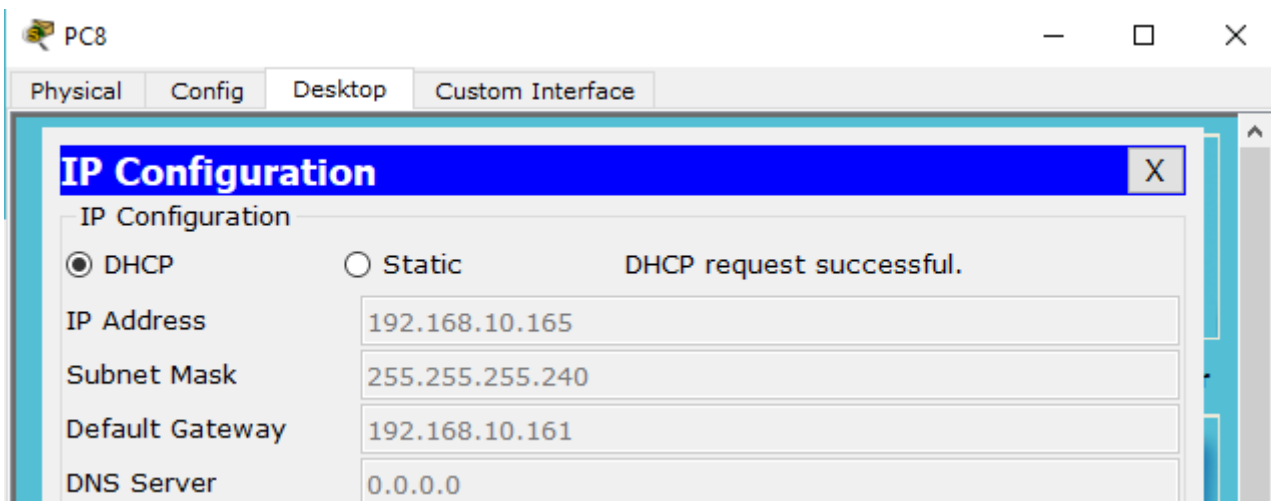
PC7:



The screenshot shows the 'IP Configuration' window for PC7. The window has tabs for 'Physical', 'Config', 'Desktop', and 'Custom Interface'. The 'Config' tab is selected. The window title is 'IP Configuration' with a close button 'X'. The 'IP Configuration' section shows 'DHCP' selected with a radio button, and 'Static' is unselected. A message 'DHCP request successful.' is displayed. The fields for IP Address, Subnet Mask, Default Gateway, and DNS Server are filled with the following values:

Field	Value
IP Address	192.168.10.150
Subnet Mask	255.255.255.240
Default Gateway	192.168.10.145
DNS Server	0.0.0.0

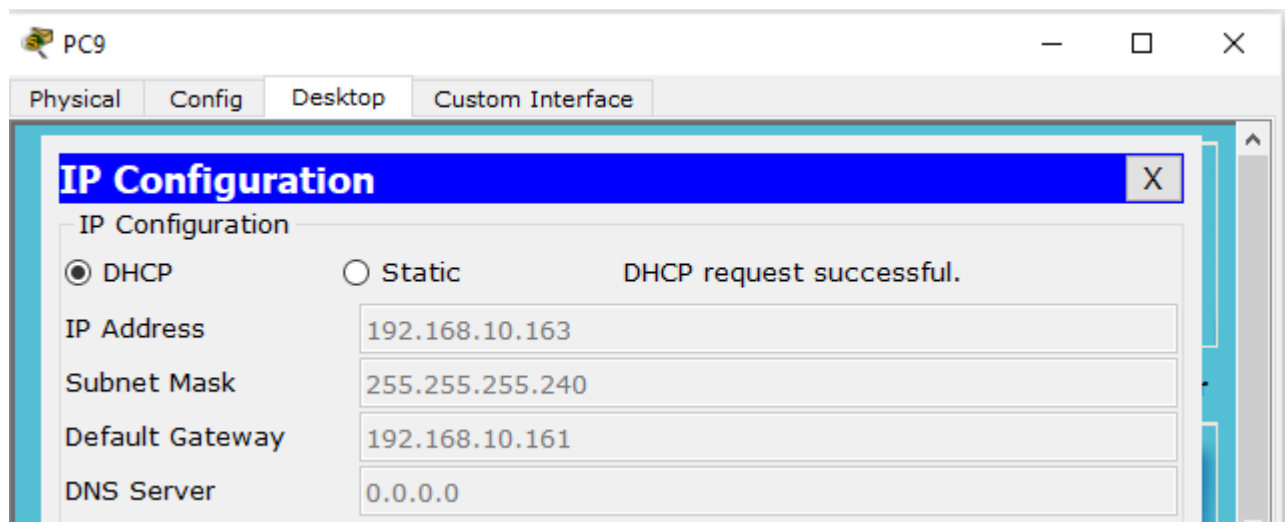
PC8:



The screenshot shows the 'IP Configuration' window for PC8. The window has tabs for 'Physical', 'Config', 'Desktop', and 'Custom Interface'. The 'Config' tab is selected. The window title is 'IP Configuration' with a close button 'X'. The 'IP Configuration' section shows 'DHCP' selected with a radio button, and 'Static' is unselected. A message 'DHCP request successful.' is displayed. The fields for IP Address, Subnet Mask, Default Gateway, and DNS Server are filled with the following values:

Field	Value
IP Address	192.168.10.165
Subnet Mask	255.255.255.240
Default Gateway	192.168.10.161
DNS Server	0.0.0.0

PC9:

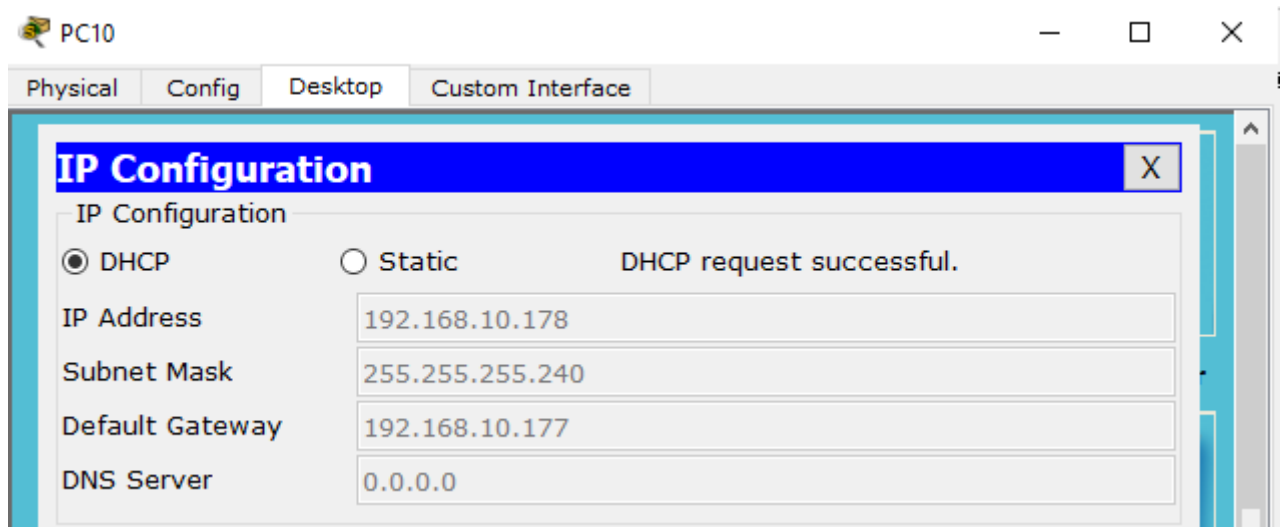


The screenshot shows the 'IP Configuration' window for PC9. The window has tabs for 'Physical', 'Config', 'Desktop', and 'Custom Interface'. The 'Config' tab is selected. The window title is 'IP Configuration' with a close button 'X'. The 'IP Configuration' section shows 'DHCP' selected with a radio button, and 'Static' is unselected. A message 'DHCP request successful.' is displayed. The fields for IP Address, Subnet Mask, Default Gateway, and DNS Server are filled with the following values:

Field	Value
IP Address	192.168.10.163
Subnet Mask	255.255.255.240
Default Gateway	192.168.10.161
DNS Server	0.0.0.0

CID: 119638

PC10:



PC10

Physical Config Desktop Custom Interface

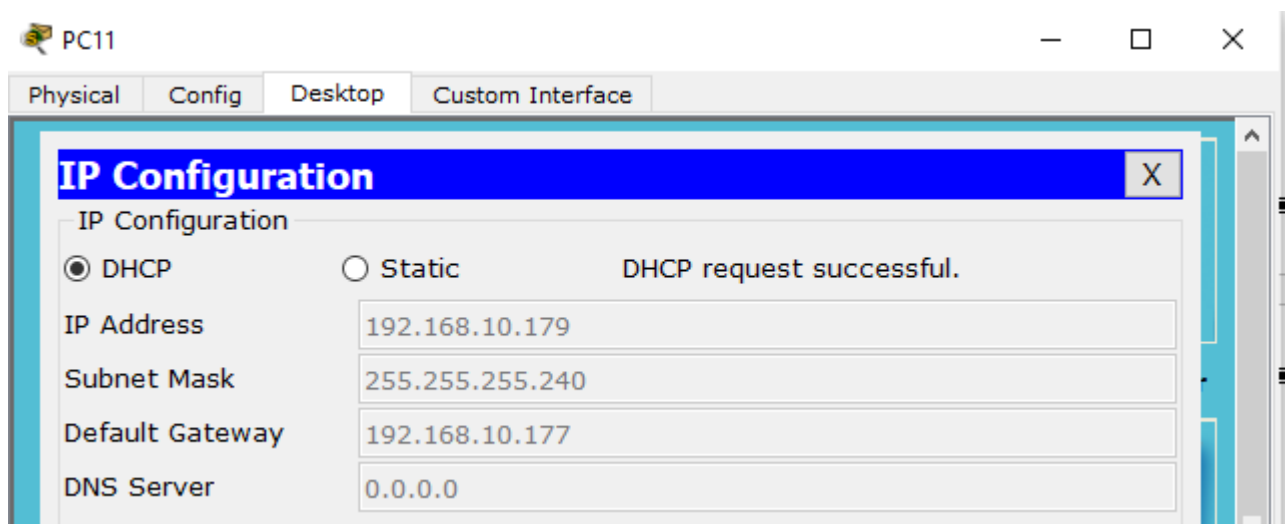
IP Configuration

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IP Address	192.168.10.178
Subnet Mask	255.255.255.240
Default Gateway	192.168.10.177
DNS Server	0.0.0.0

PC11:



PC11

Physical Config Desktop Custom Interface

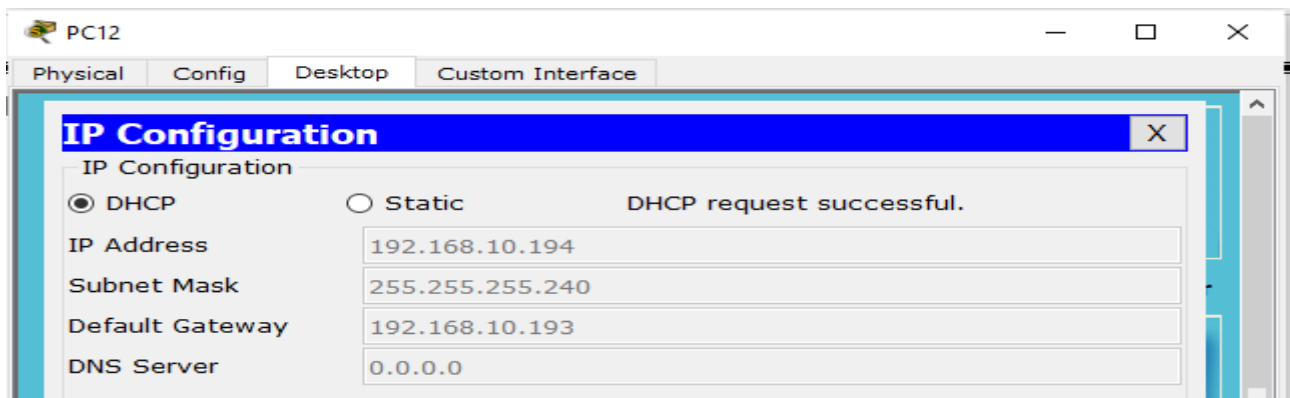
IP Configuration

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IP Address	192.168.10.179
Subnet Mask	255.255.255.240
Default Gateway	192.168.10.177
DNS Server	0.0.0.0

PC12:



PC12

Physical Config Desktop Custom Interface

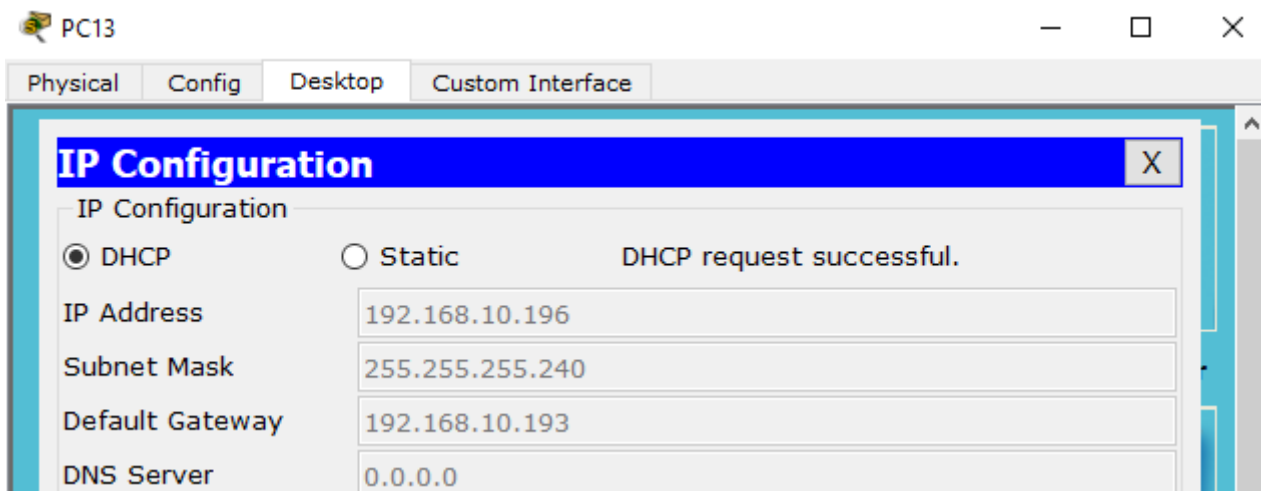
IP Configuration

IP Configuration

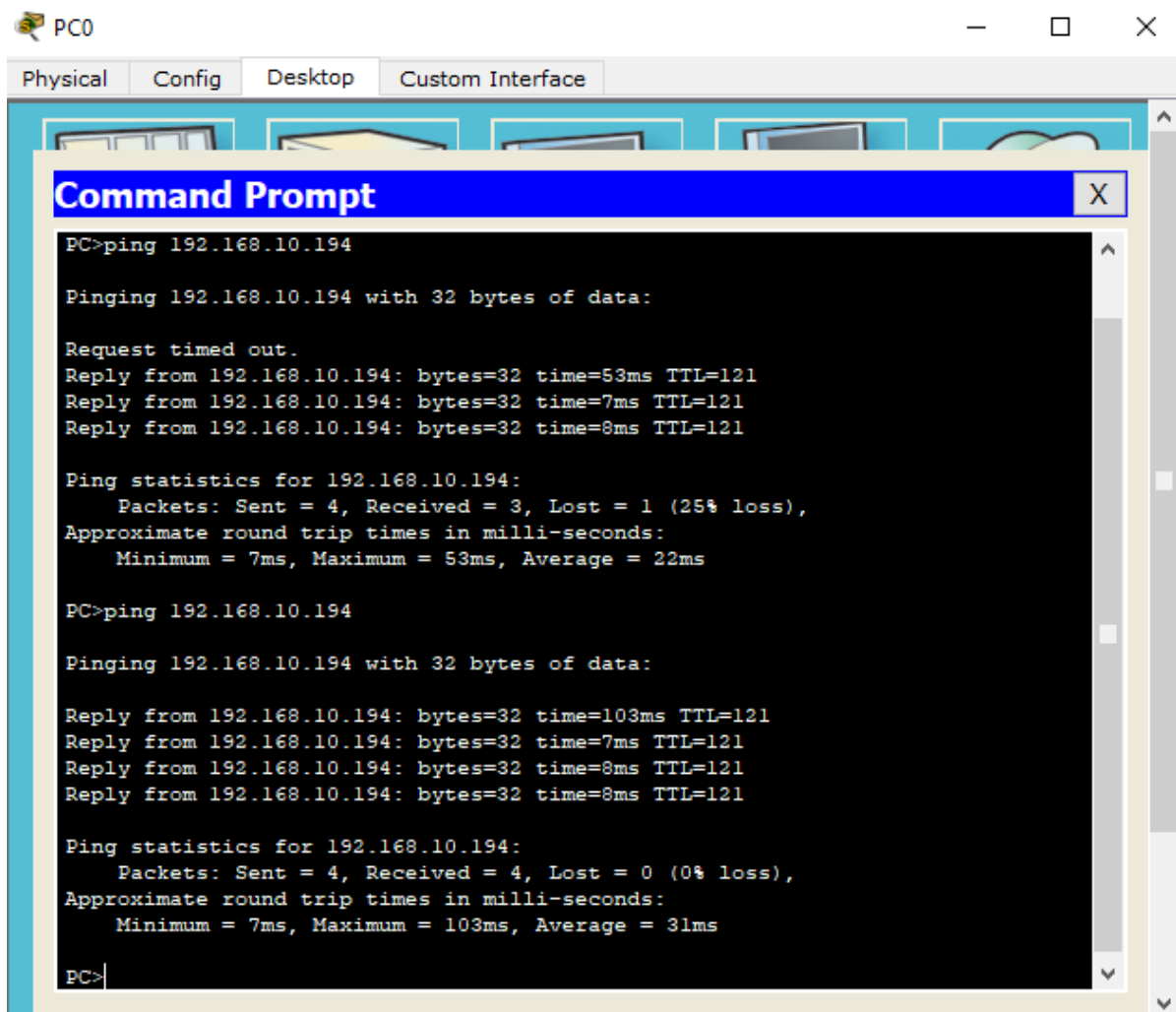
☒ DHCP ☐ Static DHCP request successful.

IP Address	192.168.10.194
Subnet Mask	255.255.255.240
Default Gateway	192.168.10.193
DNS Server	0.0.0.0

PC13:

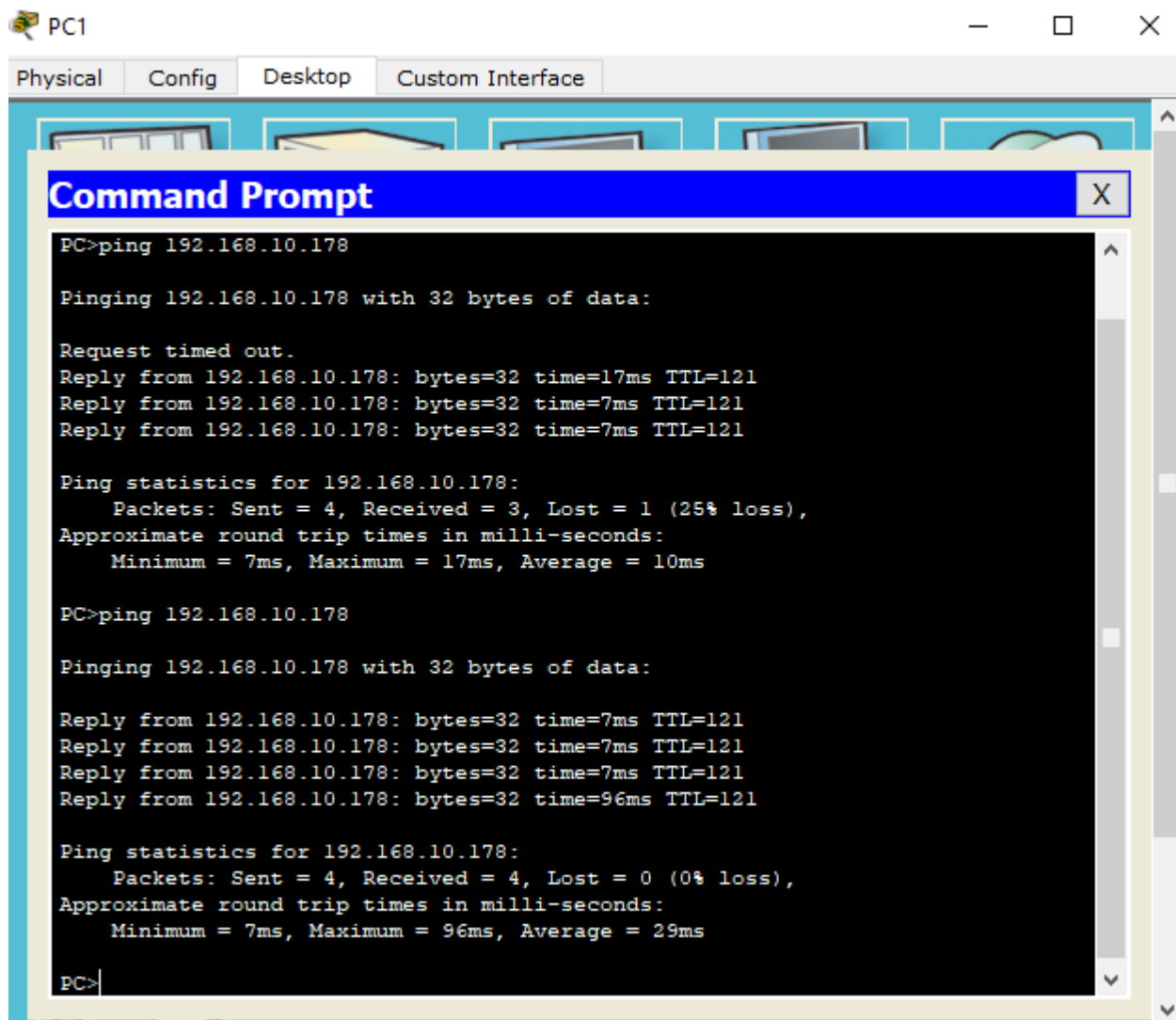


PC0:



CID: 119638

PC1:



The screenshot shows a virtual PC1 desktop with a taskbar containing icons for a folder, a document, and a network icon. The desktop background is blue with a pattern of white squares. A window titled "PC1" is open, showing a "Command Prompt" window. The Command Prompt window has a blue title bar and a black background with white text. It displays the results of two ping commands to 192.168.10.178. The first command shows a 25% loss (1 packet lost) and the second command shows 0% loss (0 packets lost).

```
PC1
Physical Config Desktop Custom Interface

Command Prompt
PC>ping 192.168.10.178

Pinging 192.168.10.178 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.178: bytes=32 time=17ms TTL=121
Reply from 192.168.10.178: bytes=32 time=7ms TTL=121
Reply from 192.168.10.178: bytes=32 time=7ms TTL=121

Ping statistics for 192.168.10.178:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 17ms, Average = 10ms

PC>ping 192.168.10.178

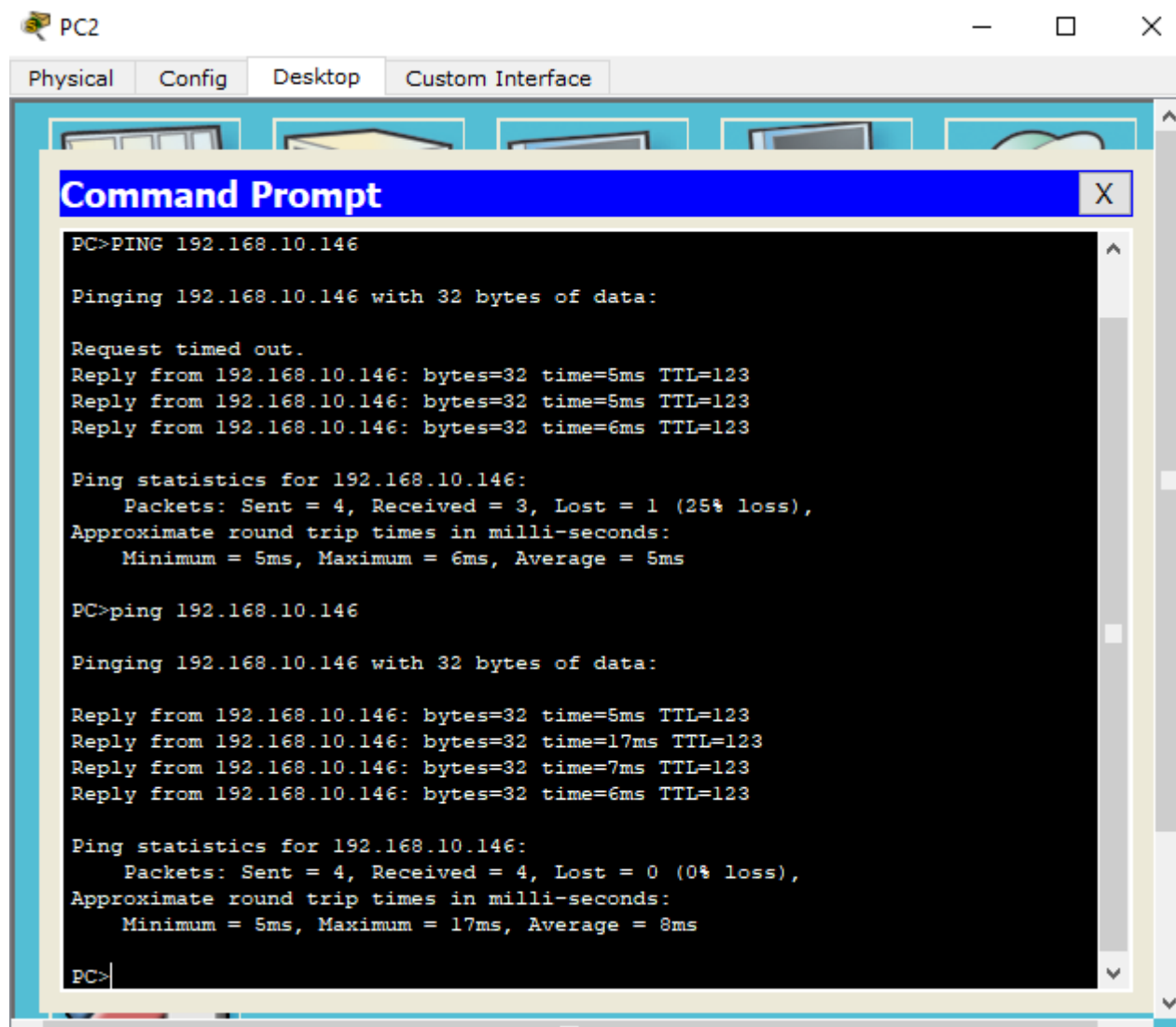
Pinging 192.168.10.178 with 32 bytes of data:

Reply from 192.168.10.178: bytes=32 time=7ms TTL=121
Reply from 192.168.10.178: bytes=32 time=7ms TTL=121
Reply from 192.168.10.178: bytes=32 time=7ms TTL=121
Reply from 192.168.10.178: bytes=32 time=96ms TTL=121

Ping statistics for 192.168.10.178:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 96ms, Average = 29ms

PC>
```

PC2:



The screenshot shows a virtual PC2 desktop with a taskbar containing icons for folders and applications. A 'Command Prompt' window is open, displaying the results of a ping command to 192.168.10.146. The window has a blue title bar and a close button. The desktop background is a light blue gradient with icons for 'My Computer', 'Network Places', 'Recycle Bin', and 'My Recent Places'.

```
PC2
Physical Config Desktop Custom Interface

Command Prompt

PC>PING 192.168.10.146

Pinging 192.168.10.146 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.146: bytes=32 time=5ms TTL=123
Reply from 192.168.10.146: bytes=32 time=5ms TTL=123
Reply from 192.168.10.146: bytes=32 time=6ms TTL=123

Ping statistics for 192.168.10.146:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 6ms, Average = 5ms

PC>ping 192.168.10.146

Pinging 192.168.10.146 with 32 bytes of data:

Reply from 192.168.10.146: bytes=32 time=5ms TTL=123
Reply from 192.168.10.146: bytes=32 time=17ms TTL=123
Reply from 192.168.10.146: bytes=32 time=7ms TTL=123
Reply from 192.168.10.146: bytes=32 time=6ms TTL=123

Ping statistics for 192.168.10.146:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 17ms, Average = 8ms

PC>
```

Hammad15398-1

 Hammad15398-1

Physical Config CLI

IOS Command Line Interface

```
Hammad15398-1>enable
Hammad15398-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Hammad15398-1(config)#interface FastEthernet0/0
Hammad15398-1(config-if)#exit
Hammad15398-1(config)#exit
Hammad15398-1#
%SYS-5-CONFIG_I: Configured from console by console

Hammad15398-1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.10.0/24 is variably subnetted, 6 subnets, 3 masks
C       192.168.10.0/27 is directly connected, FastEthernet0/0
S       192.168.10.32/27 [1/0] via 192.168.10.98
S       192.168.10.64/27 [1/0] via 192.168.10.98
C       192.168.10.96/30 is directly connected, Serial0/3/0
C       192.168.10.100/30 is directly connected, Serial0/3/1
S       192.168.10.128/25 [1/0] via 192.168.10.98
Hammad15398-1#
```

Copy

Hammad15398-2

Hammad15398-2

— □ ×

Physical Config CLI

IOS Command Line Interface

```
Hammad15398-2>en
Hammad15398-2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    192.168.10.0/24 is variably subnetted, 8 subnets, 3 masks
S       192.168.10.0/27 [1/0] via 192.168.10.97
C       192.168.10.32/27 is directly connected, FastEthernet0/0
S       192.168.10.64/27 [1/0] via 192.168.10.106
C       192.168.10.96/30 is directly connected, Serial0/3/0
C       192.168.10.100/30 is directly connected, Serial0/3/1
C       192.168.10.104/30 is directly connected, Serial0/2/0
C       192.168.10.112/30 is directly connected, Serial0/2/1
S       192.168.10.128/25 [1/0] via 192.168.10.113
Hammad15398-2#
```

Copy

Paste

Hammad15398-3

 Hamad15398-3

— □ ×

Physical Config CLI

IOS Command Line Interface

```
Hamad15398-3>en
Hamad15398-3#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.10.105 to network 0.0.0.0

    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.64/27 is directly connected, FastEthernet0/0
C       192.168.10.104/30 is directly connected, Serial0/2/0
S*    0.0.0.0/0 [1/0] via 192.168.10.105
Hamad15398-3#
```

Copy Paste

Hammad15398-4

```

Hammad15398-4>en
Hammad15398-4#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    102.0.0.0/30 is subnetted, 1 subnets
C       102.168.10.112 is directly connected, Serial0/3/1
    192.168.10.0/24 is variably subnetted, 4 subnets, 2 masks
S       192.168.10.0/25 [1/0] via 192.168.10.114
C       192.168.10.112/30 is directly connected, Serial0/2/1
C       192.168.10.116/30 is directly connected, Serial0/3/0
S       192.168.10.128/25 [1/0] via 192.168.10.114
           [1/0] via 192.168.10.118
Hammad15398-4#

```

Hammadullah15398

```

% Invalid input detected at '^' marker.

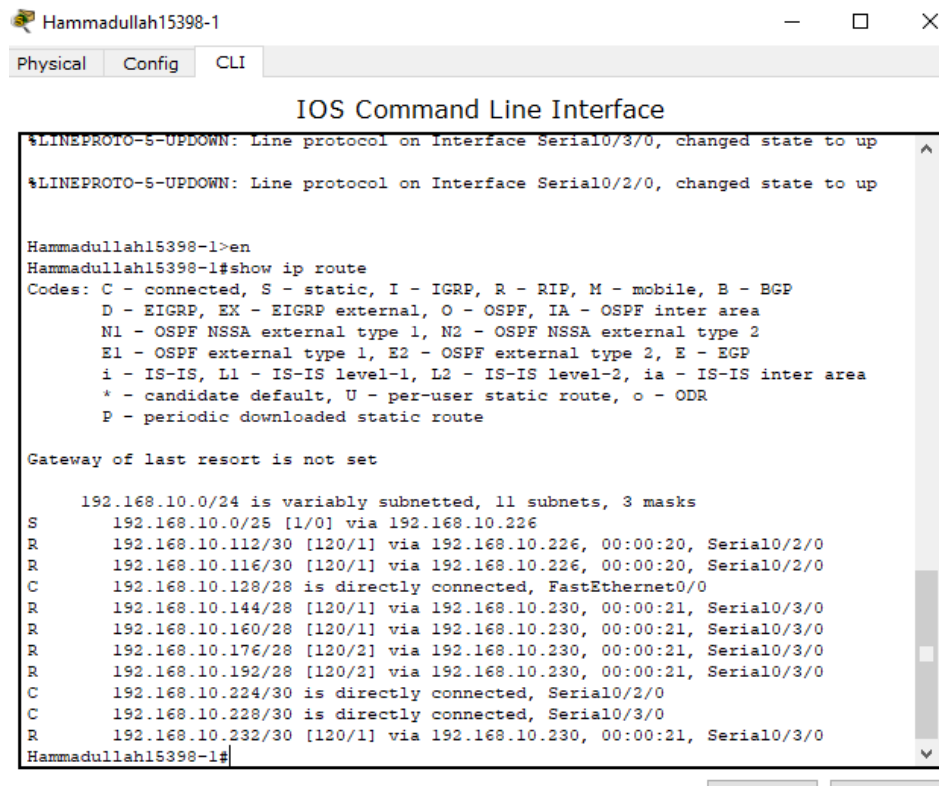
Hammadullah15398(config)#exit
Hammadullah15398#
%SYS-5-CONFIG_I: Configured from console by console

Hammadullah15398#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.10.0/24 is variably subnetted, 11 subnets, 3 masks
S       192.168.10.0/25 [1/0] via 192.168.10.117
C       192.168.10.112/30 is directly connected, Serial0/3/1
C       192.168.10.116/30 is directly connected, Serial0/3/0
R       192.168.10.128/28 [120/1] via 192.168.10.225, 00:00:07, Serial0/2/0
R       192.168.10.144/28 [120/2] via 192.168.10.225, 00:00:07, Serial0/2/0
R       192.168.10.160/28 [120/2] via 192.168.10.225, 00:00:07, Serial0/2/0
R       192.168.10.176/28 [120/3] via 192.168.10.225, 00:00:07, Serial0/2/0
R       192.168.10.192/28 [120/3] via 192.168.10.225, 00:00:07, Serial0/2/0
C       192.168.10.224/30 is directly connected, Serial0/2/0
R       192.168.10.228/30 [120/1] via 192.168.10.225, 00:00:07, Serial0/2/0
R       192.168.10.232/30 [120/2] via 192.168.10.225, 00:00:07, Serial0/2/0
Hammadullah15398#

```



Hamadullah15398-1

Physical Config CLI

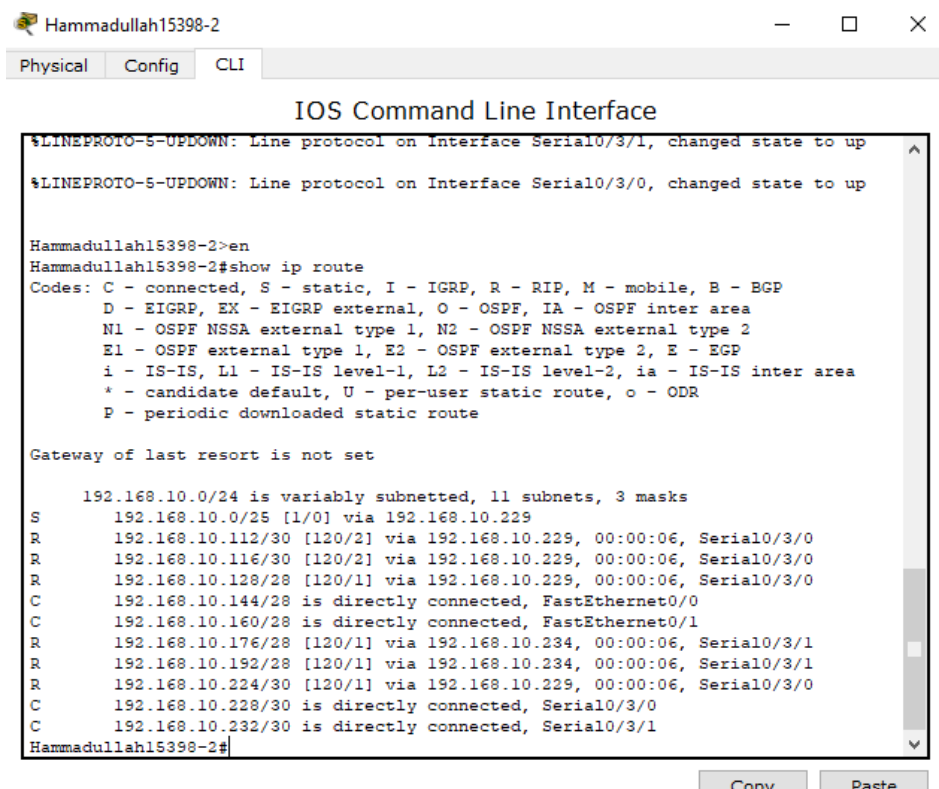
IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed state to up

Hamadullah15398-1>en
Hamadullah15398-1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

      192.168.10.0/24 is variably subnetted, 11 subnets, 3 masks
S       192.168.10.0/25 [1/0] via 192.168.10.226
R       192.168.10.112/30 [120/1] via 192.168.10.226, 00:00:20, Serial0/2/0
R       192.168.10.116/30 [120/1] via 192.168.10.226, 00:00:20, Serial0/2/0
C       192.168.10.128/28 is directly connected, FastEthernet0/0
R       192.168.10.144/28 [120/1] via 192.168.10.230, 00:00:21, Serial0/3/0
R       192.168.10.160/28 [120/1] via 192.168.10.230, 00:00:21, Serial0/3/0
R       192.168.10.176/28 [120/2] via 192.168.10.230, 00:00:21, Serial0/3/0
R       192.168.10.192/28 [120/2] via 192.168.10.230, 00:00:21, Serial0/3/0
C       192.168.10.224/30 is directly connected, Serial0/2/0
C       192.168.10.228/30 is directly connected, Serial0/3/0
R       192.168.10.232/30 [120/1] via 192.168.10.230, 00:00:21, Serial0/3/0
Hamadullah15398-1#
```



Hamadullah15398-2

Physical Config CLI

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Hamadullah15398-2>en
Hamadullah15398-2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

      192.168.10.0/24 is variably subnetted, 11 subnets, 3 masks
S       192.168.10.0/25 [1/0] via 192.168.10.229
R       192.168.10.112/30 [120/2] via 192.168.10.229, 00:00:06, Serial0/3/0
R       192.168.10.116/30 [120/2] via 192.168.10.229, 00:00:06, Serial0/3/0
R       192.168.10.128/28 [120/1] via 192.168.10.229, 00:00:06, Serial0/3/0
C       192.168.10.144/28 is directly connected, FastEthernet0/0
C       192.168.10.160/28 is directly connected, FastEthernet0/1
R       192.168.10.176/28 [120/1] via 192.168.10.234, 00:00:06, Serial0/3/1
R       192.168.10.192/28 [120/1] via 192.168.10.234, 00:00:06, Serial0/3/1
R       192.168.10.224/30 [120/1] via 192.168.10.229, 00:00:06, Serial0/3/0
C       192.168.10.228/30 is directly connected, Serial0/3/0
C       192.168.10.232/30 is directly connected, Serial0/3/1
Hamadullah15398-2#
```

Copy Paste

Hammadullah15398-3

 Hamadullah15398-3

Physical Config CLI

IOS Command Line Interface

```
up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up

Hamadullah15398-3>en
Hamadullah15398-3#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

192.168.10.0/24 is variably subnetted, 11 subnets, 3 masks
S    192.168.10.0/25 [1/0] via 192.168.10.233
R    192.168.10.112/30 [120/3] via 192.168.10.233, 00:00:20, Serial0/3/1
R    192.168.10.116/30 [120/3] via 192.168.10.233, 00:00:20, Serial0/3/1
R    192.168.10.128/28 [120/2] via 192.168.10.233, 00:00:20, Serial0/3/1
R    192.168.10.144/28 [120/1] via 192.168.10.233, 00:00:20, Serial0/3/1
R    192.168.10.160/28 [120/1] via 192.168.10.233, 00:00:20, Serial0/3/1
C    192.168.10.176/28 is directly connected, FastEthernet0/0
C    192.168.10.192/28 is directly connected, FastEthernet0/1
R    192.168.10.224/30 [120/2] via 192.168.10.233, 00:00:20, Serial0/3/1
R    192.168.10.228/30 [120/1] via 192.168.10.233, 00:00:20, Serial0/3/1
C    192.168.10.232/30 is directly connected, Serial0/3/1
Hamadullah15398-3#
```

Copy Paste